INCH-POUND

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MILITARY SPECIFICATION

TRANSMITTER, PRESSURE TRU-80/A, OIL, ENGINE-MOUNTED, 0-100 PSI

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of remote-indicating 320° movement, synchro style, differential, engine-mounted oil-pressure transmitter, designated TRU-80/A.

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEOR, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

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SPECIFICATIONS

FEDERAL

QQ-S-763	Steel Bars, Shapes, And Forgings-Corrosion Resisting
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods Of
MIL-E-5400	Electronic Equipment, Airborne, General Specification For
MIL-L-7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Number O-148
MIL-C-5541	Chemical Conversion Coatings On Aluminum And Aluminum Alloys
MIL-I-7057	Indicators, Synchro, Aircraft, General Specification For
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-S-20708	Synchros, General Specification For
MIL-B-27497	Bearings, Jewel, Sapphire Or Ruby, Synthetic

STANDARDS

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements For Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization Of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MIL-STD-970	Standards & Specifications, Order Of Preference For The Selection Of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, And Waterproofing, With Appropriate Test Methods
MIL-STD-2073/1A	DOD Material, Procedure or Development And Application Of Packaging Requirements
MS33649	Bosses, Fluid Connection - Internal Straight Thread
MS33678	Connector, Receptacle, Electric, Integral Mounting

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HANDBOOKS

MILITARY

MIL-HDBK-275	Guide For Selection Of Lubrications, Fluids, And Compounds For Use In Flight Vehicles And Components

PUBLICATIONS

MILITARY

SD-6	Provisions Governing Qualification

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Transmitter furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.4 and 6.3).

3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.3 Materials. Reclaimed materials shall be utilized to the maximum extent possible within the quality limits required by this document. Materials used in the manufacture of transmitters shall be suitable for the purpose and shall conform to

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applicable Government specifications. Materials conforming to contractor's specifications may be used provided the specifications are approved by the Government and contain provisions for adequate tests. The use of the contractor's specification will not constitute waiver of Government inspection.

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the contracting activity. However, if they are used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.2 Metals. Metals shall be corrosion-resistant or suitably treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met in storage or normal service. Cadmium shall not be used without the approval of the preparing activity.

3.3.3 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.4 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the transmitter, except where magnetic materials are essential.

3.3.5 Corrosive fumes. Materials used in the construction of the transmitter shall not produce corrosive or deleterious fumes under the test conditions specified herein.

3.3.6 Protective treatment. When materials are used in the construction of the transmitter that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction.

3.4.1 Design. The transmitter shall be designed to operate with indicators conforming to MIL-I-7057.

3.4.1.1 Outline dimensions. The outline dimensions of the transmitter shall conform to figure 1.

3.4.1.2 Adjustments. The transmitter shall be provided with a means for adjusting or correcting indications which shall be simple to manipulate and such that adjustments can be accomplished with tools ordinarily available. The means of adjustment shall be accessible from the exterior of the case and shall provide a minimum of ± 7 angular degrees of synchro rotor adjustment from electrical zero.

3.4.1.3 Mechanisms. Transmitter mechanisms shall be comprised of units distinct from their cases. The units shall be sufficiently rugged so that small distortions of the cases will not affect the accuracy of the indications.

3.4.1.4 Adjustment locks. All calibration adjustments shall be provided with securing devices to lock the adjusting mechanism in place to prevent the adjustment from drifting from its original setting due to environmental conditions specified

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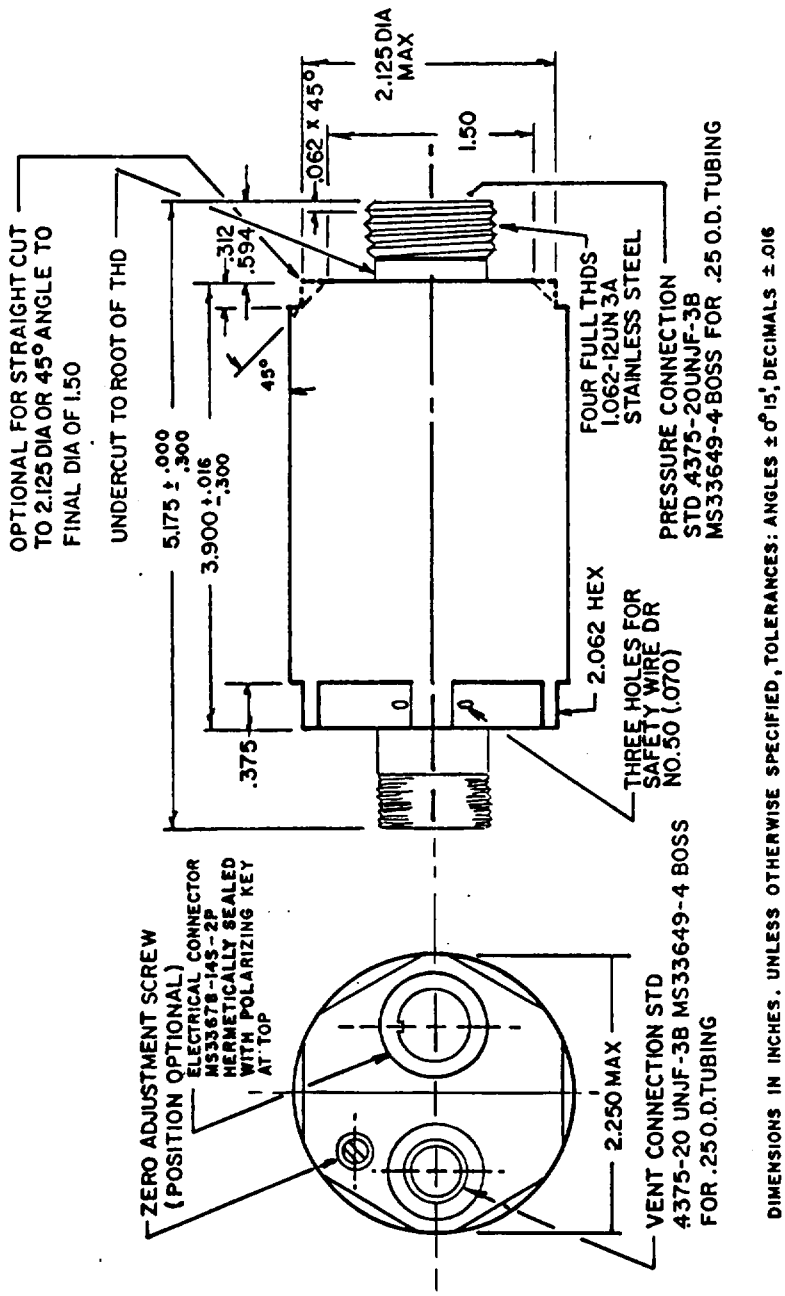


FIGURE 1. Transmitter case and connectors.

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herein. Provisions for locking shall be made on the calibration adjustment so that it cannot be readily moved.

3.4.1.5 Reliability. The transmitter shall have a specified meantime between failures (MTBF) of 2,000 hours at a 90 percent confidence level.

3.4.1.6 Lubricant. The contractor shall select a lubricant or lubricants in accordance with MIL-HDBK-275 which will be suitable for use on gears, joints, bearings, and pinions of the transmitter. Prior to selecting the lubricant or lubricants, the contractor shall make an investigation relative to the lubricant most suitable for the applications with due consideration being given to the environmental conditions under which the transmitter must operate. The type of lubricant and the reason for its selection shall be specified by the contractor and approved by the contracting activity.

3.4.1.7 Synchro transmitter. Each transmitter shall be capable of operating either one or two indicators, whose synchros conform to the electrical characteristics of MIL-S-20708/5, without change in indication of more than one degree of dial arc.

3.4.1.7.1 Synchro bearings. The precision ball bearings used in the synchro shall be corrosion-resistant alloy steel (stainless steel) conforming to QQ-S-763, class 440C, and shall be made to ABEC tolerance 5P.

3.4.1.7.2 Brushless synchro. The brush and slip ring method shall not be used in the synchro transmitter to transport excitation power to the synchro rotor. The excitation power shall be transported by means of a continuous electrical circuit which shall permit the rotor to turn readily through a arc of 340° minimum. Stops shall be provided to prevent overwinding. A means shall be provided for readily setting the synchro at electrical zero. The synchro transmitter design shall be such that the synchro electrical zero position can be set and locked in position so securely that it will remain in that position for the service life of the transmitter or until reset.

3.4.1.8 Overpressure stop. An overpressure stop shall be provided to restrain the prime mover. The stop shall permit the transmitter rotor to rotate 12° +4.5° beyond the 100-psi position. The stop shall not be damaged by application of 500 psi overpressure.

3.4.1.9 Gears, pivots, joints, bearings. Before finalizing the design of the pressure transmitter mechanism, the contractor shall investigate and determine the best known types and the proper processing of interfacing materials for use in the gears, pivots, joints, and bearings in the train of moving parts of the transmitter. These interface materials shall be selected for their ability to resist wear, corrosion, and friction when interfacing with one another. The interfacing materials shall not be dissimilar as described in MIL-STD-889. After selection of the materials types and processing procedures, the contractor shall submit a description of each interfacing material and process proposed for the train of moving parts (gears, pivots, joints, bearings) and present his reason for selection of each type to the contracting activity or approval.

3.4.1.9.1 Large-area interface surface. The gears, pivots, joints, and bearings shall have substantially large interface surfaces for the purpose of minimizing the effects of wear upon the calibration and accuracy of the transmitter. Single point type interfaces, such as attained by pivots made up of two perpendicular cylinders will not be acceptable, while interface surfaces made up of two parallel cylinders or curved surfaces will be acceptable if sufficient interface area is provided to

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minimize the effects of wear. The contractor shall furnish explicit details in his engineering report concerning this design factor and shall show how his proposed design will minimize wear.

3.4.1.10 Attitude. The transmitter shall be so designed that it will operate as specified herein in any position.

3.4.1.11 Differential pressure. The differential pressure transmitter shall have two connections (pressure connection and vent connection) as shown in figure 1. The transmitter shall be so designed and constructed that it will be actuated by the difference between the pressure applied to the pressure connection and the pressure applied to the vent connection. The synchro signal output shall represent pressure connection pressure minus vent connection pressure. The angular position of the rotor of the synchro relative to the stator shall be essentially a linear function of pressure minus vent connection pressure.

3.4.1.12 Negative pressure. The transmitter shall be constructed to withstand application of a differential pressure (vent connection pressure greater than pressure connection pressure) of 25 psi without affecting accuracy.

3.4.2 Construction. The transmitter shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipment, storage, installation, and service. Pivots, bearings, and gears shall not bind nor shake and shall be as nearly frictionless as possible.

3.4.2.1 Mounting torque. The transmitter shall be of such rigid construction that it will withstand a torque of 720 inch-pounds applied to the 2.062-inch hex nut (see figure 1) without distorting the case or affecting the accuracy.

3.5 Performance. The transmitter shall be designed and calibrated to operate continuously and produce its greatest accuracy at a normal operating temperature of 177°C. The transmitter shall withstand and operate while being subjected to the following conditions:

a. Temperatures: Operating temperatures ranging from -54° to +177°C and exposure to temperatures ranging from -62° to +232°C.

b. Vibration: Random vibration of 25 GRMS.

c. Altitude-temperature: Absolute pressure equivalent to an 85,000-foot altitude at 54°C.

d. Salt fog: Exposure to salt sea atmosphere for 48 hours.

e. Overpressure: An overpressure of 200 psi for a 10-minute period.

f. Humidity: Exposure to humidity for 240 hours (non-hermetically sealed transmitters only).

3.5.1 Dielectric strength. The transmitter shall withstand 200V at a commercial frequency applied between any pin of the electrical connector, not connected to the case, and the case for minimum period of 5 seconds.

3.5.2 Voltage and frequency variation. The transmitter shall withstand voltage and frequency variations from 22 to 30V and 320 to 480 Hz or any combinations thereof.

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3.5.3 Electromagnetic interference. The transmitter shall meet the electromagnetic interference requirements of MIL-STD-461 as specified for class A1 equipment.

3.5.4 Damping. The transmitter output response to a 100 to 0 psi pressure change shall be 100 to 10 psi in 4±2 seconds at -40°C. The response shall be 100 to 10 psi in 0.1 to 1.0 seconds at 177°C. At -40°C the response to a 0 to 50 psi pressure change shall be 0 to 40 psi in at least 22 seconds.

3.5.5 Indicator operation. The transmitter shall be capable of operating either one or two indicators without a change in output equivalent to more than 1° of arc.

3.5.6 Vent boss and connector. The vent boss and connector shall withstand and applied force of 25 foot-pounds bending moment with no evidence of damage to welds or soldering, distortion of the base plate, and without the calibration shifting out of tolerance.

3.5.7 Scale error. Scale error shall not exceed the tolerances specified in table I.

TABLE I. Scale error tolerances (±PSI.)

Test Pressure PSI	Room Temp, High Temp, Normal Operating Temp (177°C), Humidity, Mounting Torque	AFTER: Overpressure High Temp Exposure, Vibration Reliability	DURING: Low Temp, Low Temp Exposure, High Temp Exposure, Sensors Matching
0	2.0	3.0	4.0
20	2.0	3.0	4.0
30	1.5	2.5	3.0
40	1.5	2.0	2.5
45	1.5	2.0	2.5
50	1.5	2.0	2.5
60	1.5	2.0	2.5
70	1.5	2.5	3.0
80	2.0	3.0	4.0
100	2.0	3.0	4.0

3.5.8 Friction and position error. Friction and position error shall not exceed the tolerances specified in table II.

TABLE II. Friction and position error.

Tests	Tolerances (±PSI)
Friction:	
High Temperature	2.0
High Temperature Exposure	2.0
Low Temperature	2.0
Low Temperature Exposure	2.0
Room Temperature	2.0
Operating Temperature	2.0
Position Error at 50 PSI:	0.5
Maximum pointer oscillation	1.0
Maximum pointer variation	1.0

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3.5.9 Effect of lead length. When the lead length (No. 20 AWG stranded copper wire) between the transmitter and indicator synchro stators is changed from 5 to 300 feet, the indication shall not change more than 1° of arc.

3.5.10 Leakage

3.5.10.1 Hermetically sealed case. When a pressure of approximately 0 psia is applied to the exterior of the transmitter case, the leak rate shall not exceed 0.10 micron cubic foot per hour when 10-percent helium is used in the filling medium and shall not exceed 1.0 micron cubic foot per hour when 100 percent helium is used as the filling medium.

3.5.10.2 Nonhermetically sealed case. There shall be no leakage when the transmitter is submerged in a partially filled container of water which has been evacuated to a air pressure of 1.0 psi.

3.5.11 Differential pressure. The differential pressure transmitter shall be capable of receiving input pressures varying from 0 to 100 PSI applied to the pressure connection in excess of the vent pressure and input pressures varying from 0 PSI to 40 PSI applied to the vent connection in any pressure combinations as long as the input pressure exceeds the vent pressure. Scale error shall not exceed the tolerances specified in Table I for "Sensors Matching" at room temperature.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-STD-100 shall govern the manufacturer's part number and changes thereto.

3.7 Case. The case shall conform to figure 1 and shall provide either a hermetically sealed or gasket sealed enclosure for the internal mechanism of the transmitter. Sealing of the case shall be so accomplished that the seal will not be dependent upon materials which would be affected by the action of any atmosphere to which the transmitter may be subjected.

3.7.1 Filling medium. When a hermetically sealed case is used, the filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be either a mixture of 88 to 92 percent nitrogen and the remainder helium or 100 percent helium. The absolute pressure of the filling medium in the case shall be 12 to 14.7 psi.

a. The transmitter shall not contain silicone oil.

3.8 Electronic components. Electronic components and the application thereof shall be in accordance with MIL-E-5400. Nonstandard parts shall not be used except with prior approval of the contracting activity.

3.9 Electrical connector. An MS33678-14S-2P hermetically sealed electrical connector shall be provided.

3.10 Electrical characteristics. The transmitter synchro shall meet the

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requirements of 3.10.1 through 3.10.3 with a normal excitation of 26V 400-Hz single-phase power applied to the primary (rotor) from a source having electrical characteristics conforming to MIL-STD-704 as specified for category B equipment.

3.10.1 Open secondary. The current drawn by the primary of the self-synchronous motor shall not exceed 240 ma with second-aries (stator) open circuited.

3.10.2 Power input. Power consumption of the transmitter synchro shall not exceed 6.24 va.

3.10.2 Secondary voltage. The maximum open-circuit voltage measured between any two secondary leads shall be 11.8 ±0.3V. The variation between the three maximum secondary voltages on any one synchro shall not exceed 0.2V.

3.11 Wiring.

3.11.1 External. The electrical receptacle shall be so wired that the 26-V, 400-Hz power supply will be connected to terminals A and B. External wiring between the transmitter and indicator shall be as shown on figure 2.

3.11.2 Internal. Internal wiring of the transmitter shall be color coded and shall be connected to the receptacle pins as shown on figure 3. When gears are used to drive the transmitter counterclockwise for increasing values of the parameter being measured, terminals S1 and S3 shall be reversed internally to cancel the effects of the gear reversal to assure clockwise rotation of the indicator synchro shaft and pointer with increasing values.

3.12 Jewel bearings. When used, synthetic sapphire or ruby jewel bearings shall be in accordance with MIL-B-27497.

3.13 Finishes and protective coatings.

3.13.1 Aluminum alloy parts. Aluminum alloy parts shall be coated with an anodic film conforming to MIL-A-8625 except that small holes, straight threads, case inserts, and parts in hermetically sealed enclosures need not be anodized.

a. Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film conforming to MIL-C-5541.

b. When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.14 Weight. The weight of the transmitter shall not exceed 1.75 pounds.

3.15 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.16 Identification of product. Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.16.1 Nameplate. A nameplate shall be permanently attached to the rear of the case or the required information shall be permanently etched on the side of the case where paint is not needed as a protective coating. Where applicable, the information marked thereon shall conform to MIL-STD-130. The nameplate shall not be glued or cemented to the case nor shall metals decals be used. Etching shall be permanently inscribed so that it will withstand all normal environments for the life of the transmitter.

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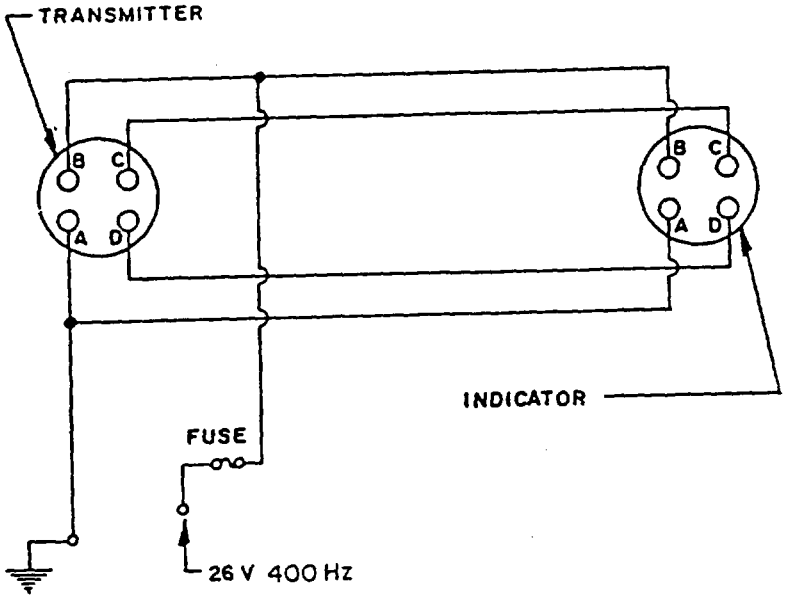
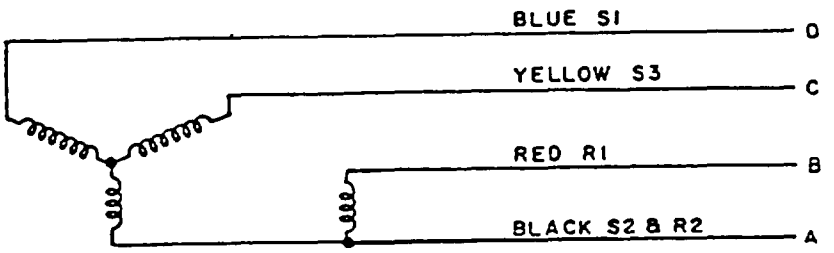


FIGURE 2. Wiring diagram - external, single indicator.

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SHOWN FOR INCREASING PRESSURE = CW SYNCHRO ROTATION. (IF INCREASING PRESSURE = CCW SYNCHRO ROTATION, THEN EXCHANGE BLUE S1 WITH YELLOW S3.)

FIGURE 3. Internal wiring - transmitter.

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3.17 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9.

3.17.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances affect the interchangeability, operation, or performance of the transmitter, they shall be held or limited accordingly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in paragraph 4.3.1 of this specification.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, the test shall be conducted at an atmospheric pressure of approximately 29.92 inches Hg and at an ambient temperature of 177°±5°C. For tests conducted at room temperature, the temperature shall be 25° ±5°C.

4.3.2 Test voltage. The transmitter shall be tested with 26V 400-Hz single-phase power. Unless otherwise specified, the voltage and frequency shall be maintained within ±1 percent of the nominal value.

4.3.3 Tapping. The transmitter shall be lightly tapped or vibrated before each scale error test reading. If vibration is used instead of tapping, then it shall be at a frequency of 60 ±10 Hz with a double amplitude less than 0.002 inch.

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4.3.4 Test standard. The transmitter shall be tested against an indicating standard having a maximum error of ±0.1 of a circular degree. The transmitter shall be tested using a pressure source which shall have a maximum inherent error of ±0.03 psi. A suitable gas may be used as the pressure medium for scale error and friction tests.

4.3.5 Adjustment. Synchro rotor adjustment shall not be used to bring scale within tolerances after tests are initiated except that when a failure is charged to a scale error shift during reliability testing, an adjustment may be made.

4.4 Qualification testing.

4.4.1 Test samples. The test samples consist of a minimum of 15 transmitters representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the contracting activity.

4.4.1.1 Disposition of samples upon completion of tests. Any sample tested may be delivered on contract, provided it meets the following:

a. It is representative of production units currently being accepted.

b. It is in good-as-new condition or has been refurbished (see 6.4.1 and 6.4.2).

c. It is otherwise satisfactory.

4.4.2 Qualification inspection. The qualification inspections shall consist of all the tests specified under 4.6.

4.5 Quality conformance inspections. Quality conformance inspections shall consist of:

a. Individual tests.....See 4.5.1

b. Sampling tests.....See 4.5.2

4.5.1 Individual tests. Each transmitter shall be subjected to the following tests as described under 4.6:

a. Examination of product.....See 4.6.1

b. Scale error test.....See 4.6.2

c. Friction error test.....See 4.6.3

d. Position error test.....See 4.6.4

e. Dielectric strength test.....See 4.6.5

f. Leakage tests.....See 4.6.6.1 and 4.6.6.2

g. Sensors matching (20 PSI Vent Pressure).....See 4.6.26.1

4.5.2 Sampling tests. One transmitter shall be selected at random from each one hundred or less produced on the contract or order and subjected to the following tests as described under 4.6:

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a. Individual tests.....See 4.5.1

b. Low temperature test.....See 4.6.8

c. High temperature test.....See 4.6.9

d. Vibration test.....See 4.6.10

e. Electrical characteristics.....See 4.6.11

f. Sensors matching.....See 4.6.26

4.5.2.1 Rejection and retest. When one item selected from a production run fails to meet the specification, transmitters on hand or later produced shall not be accepted until the extent and cause of failure have been determined.

4.5.2.2 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of transmitters on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in transmitters already accepted. If so, the contractor shall fully advise the contracting activity of all defects likely to be found and methods of correcting them.

4.6 Test methods.

4.6.1 Examination of product. The transmitter shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, marking, dimensions, and weight.

4.6.2 Scale error test. The transmitter shall be connected to an indicating standard. The transmitter shall be tested for scale error at both room and normal operating temperature at the pressures listed in table 1. These test pressures shall be applied to the transmitter pressure connection with the vent connection open to room ambient pressure. The readings of the indicating standard for both upscale and downscale shall be recorded at each test pressure. Upscale readings shall be recorded after bringing the pressure up to, but not exceeding, the specified test pressure and then tapping the transmitter. The difference between the reading and the applied pressure shall not exceed the tolerances specified in table 1. Downscale readings shall be measured in the same manner, except that the pressure shall be brought down to, but not below, the specified test pressure.

4.6.3 Friction error test. The friction error test shall be conducted at both normal operating and room temperatures and may be performed concurrently with the scale error test. The difference in readings before tapping the transmitter and after tapping has been applied shall be the friction error which shall not exceed the tolerances specified in table II.

4.6.4 Position error test. The position error test shall may be conducted in conjunction with the test specified in 4.6.2, except at room temperature only. The transmitter shall be energized to obtain a reading of approximately one-half scale. The change in indication produced by rotating the transmitter from the normal operating position to a position 90° clockwise and counterclockwise (longitudinal

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axis vertical) shall be noted. The maximum change in indication shall not exceed the tolerance specified for position error in table II.

4.6.5 Dielectric strength test. This test shall be conducted at room temperature. A potential of 200 V rms at a commercial frequency shall be applied between each pin of the electrical connector, not connected to the case, and the case for a minimum period of 5 seconds. There shall be no insulation breakdown.

4.6.6 Leakage tests.

4.6.6.1 Hermetically sealed case. The hermetically sealed transmitter shall be tested for leaks at room temperature by means of a mass-spectrometer-type helium leak detector. The detected leak rate for the 10-percent helium and 90-percent nitrogen filling medium shall not exceed 0.1 micron cubic foot per hour and for the 100-percent-helium filling medium shall not exceed 1.0 micron cubic foot per hour at an absolute pressure differential of 12 to 14.7 psi.

4.6.6.2 Nonhermetically sealed case. The nonhermetically sealed transmitter shall be immersed in a container which shall be filled to not more than 70 percent of its capacity. The water level shall be 2 to 4 inches above the nearest part of the transmitter housing. The container shall first be evacuated to 0.50 psi air pressure and held at this pressure for 1 minute or until air bubbles cease to be given off by the liquid, whichever is greater. The pressure shall then be raised to 1.0 psi and held at this pressure for a minimum period of 5 minutes. Any bubbles resulting from air or fluid leakage from within the case shall be cause for rejection. Bubbles that are the result of entrapped air on the various exterior parts of the case shall not be considered as leaks.

4.6.6.3 Pressure element. Puncture the pressure sensing element or its connection to the transmitter pressure port. Apply MIL-L-7808 oil at 150 psi to the transmitter pressure port. After 30 minutes, there shall be no evidence of leakage from the transmitter.

4.6.7 Electrical zero and verification of terminal leads. This test shall be conducted at room temperature. With power applied to the transmitter per figure 3, pressure shall be applied at the pressure connection of the transmitter with the vent connection open to ambient pressure. Electrical zero and verification of terminal leads shall be as follows:

a. The voltage from S3 to S1 (S3S1) shall begin at a null, shall be in phase with rotor voltage R2R1 as the pressure is decreased from approximately 50 psig to 0 psig.

b. The voltage between R1 and S3 shall be near a minimum and shall decrease to a minimum as the pressure is increased from 50 psig to 59.37 psig.

c. Increasing pressure from 50 psig to 59.37 psig shall result in an increasing S3S2 voltage and a decreasing S2S1 voltage.

4.6.7.1 Electrical zero adjustment. With pressure applied, the transmitter shall be subjected to, and shall meet, the test specified in 4.6.7 during which the zero adjustment shall be operated to the extreme positions. The required minimum adjustment shall be ± 7 angular degrees from the electrical zero position.

4.6.8 Low temperature test. The transmitter shall be subjected to a low temperature test in accordance with MIL-STD-810, method 502.2, procedure II at -54 ± 2°C for a period of 4 hours after which it shall be subjected to the scale error

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and friction error tests while still at that temperature. The errors shall not exceed the tolerances specified in tables I and II, respectively, for low temperature.

4.6.9 High temperature test. The transmitter shall be placed in a chamber and subjected to a temperature of 177° $\pm \frac{5}{0}$°C for a 4-hour period. Throughout this test, the transmitter shall be electrically connected to an appropriate indicator with power applied. At the end of the 4-hour period and with the transmitter still at the specified temperature, the scale error and friction error tests shall be conducted. The scale error and friction error shall not exceed the tolerances specified in tables I and II.

4.6.10 Vibration test. The transmitter shall be subjected to a random vibration test in accordance with MIL-STD-810, method 514.3. The test shall be conducted using the procedures and conditions established in MIL-STD-810, Table 514.3-I for Category 4 (ENGINE MOUNTED) equipment. The vibration spectrum used for this test shall be that defined in Figure 4 of this specification. The test shall be conducted with a suitable liquid at 50±10 psi applied to the transmitter pressure port and with the transmitter subjected to a temperature of 177° ±5°C. The transmitter shall be subjected to two continuous hours of vibration along each of three orthogonal axes. Vibration shall be applied to one axis at a time for a total of six hours of excitation. During the test, the transmitter shall meet the maximum pointer oscillation and variation tolerances of Table II. Upon completion of test, the transmitter shall be subjected to and pass the tests specified in paragraphs 4.6.2, 4.6.3, and 4.6.6.1 or 4.6.6.2, as applicable.

4.6.11 Electrical characteristics. The transmitter shall be tested to determine compliance with the requirements specified herein with respect to primary current, power input, secondary voltage, and operation of either one or two indicators with a change in indication of not more than 1° of arc.

4.6.12 Low temperature exposure test. The transmitter shall be subjected to a low temperature exposure test in accordance with MIL-STD-810, method 502.2, procedure I. The transmitter shall be exposed to a temperature of -62° ±2°C for a minimum period of 48 hours after which it shall be operated throughout its entire range. It shall exhibit no erratic characteristics during operation at the -62° ±2°C temperature. The temperature shall then be raised to -54° ±2°C and the transmitter maintained at this temperature for a minimum of 24 hours. At the end of the 24-hour period, the transmitter shall be subjected to the scale error and friction error tests. The scale error and friction error shall not exceed the tolerances specified in tables I and II for low temperature exposure.

4.6.13 High temperature exposure test. The transmitter shall be placed in a chamber and subjected to a temperature of 230° ±5°C for a period of 80 hours. At the end of this period and while still at the specified temperature, the transmitter shall be subjected to the scale error test. The transmitter shall then be returned to room temperature for 4 hours after which it shall again be subjected to the scale error test. The scale errors shall not exceed the tolerances specified in table I for during and after high temperature exposure.

4.6.14 Altitude-temperature test. The transmitter shall be subjected to an absolute pressure equivalent to an altitude of 85,000 feet for a period of 1 hour at a temperature of -54° ±2°C. While still at this pressure and temperature, the transmitter shall be subjected to the scale error test. The scale error shall not exceed the tolerances specified in table I for low temperature.

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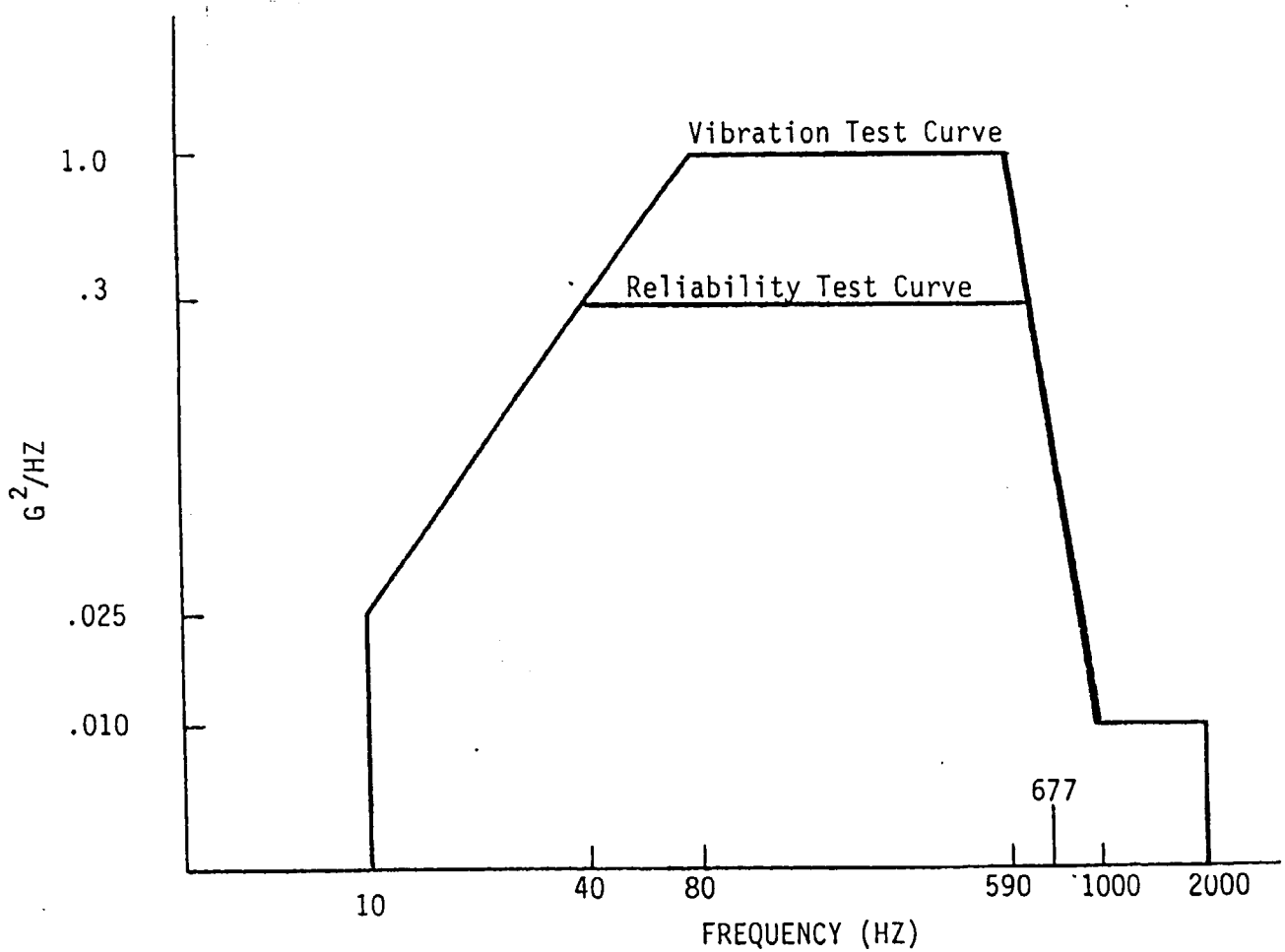


FIGURE 4. Random vibration spectrum curves

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4.6.15 Electromagnetic interference. The transmitter shall be subjected to the electromagnetic interference tests specified in MIL-STD-461 up to 400 MHz for class ID equipment. The methods used shall be as specified in MIL-STD-462.

4.6.16 Humidity test (nonhermetically sealed case). The humidity test shall be conducted in accordance with MIL-STD-810, method 507.2, procedure III. Any evidence of corrosion, other deterioration, or moisture penetration shall be cause for rejection. At the completion of this test, the transmitter shall be subjected to the scale error and friction error tests at room temperature. The errors shall not exceed the tolerances specified in table I and table II.

4.6.17 Salt-fog test. The salt-fog test shall be conducted in accordance with MIL-STD-810, method 509.2 at room temperature for a period of 48 hours. The transmitter shall be examined and there shall be no evidence of external corrosion or deterioration that would affect subsequent operation. The transmitter shall then meet the test specified in 4.6.5. Before the dielectric strength test is conducted, the external surface of the transmitter may be washed with distilled water and air dried.

4.6.18 Voltage and frequency variations. The transmitter shall be subjected to the scale error and friction error tests during which the voltage and frequency shall be varied from 22 to 30V and 320 to 480 Hz, or any combinations within these specified ranges. The scale and friction errors shall not exceed those specified in tables I and II, respectively. This test shall be conducted at room temperature.

4.6.19 Effect of lead length. The lead length between the transmitter and indicator synchro stators shall be changed from 5 to 300 feet of No. 20 AWG stranded copper wire. The indication shall not change more than 1° of arc. This test shall be conducted at room temperature.

4.6.20 Vent boss and connector strength test. The vent boss and connector mounting shall be sufficiently rigid to withstand a force of 25 foot-pounds bending moment applied in any direction perpendicular to the longitudinal axis of the transmitter without the calibration shifting out of tolerance, distortion of the base plate, or evidence of damage to the vent boss, connector welds, or soldering. This test shall be conducted with the transmitter secured by the engine mount connection or by the adjacent hex nut only. The transmitter shall then be subjected to and shall meet the paragraphs 4.6.6.1 and 4.6.6.2, as applicable. This test shall be conducted at room temperature.

4.6.21 Overpressure tests.

4.6.21.1 Overpressure (200 psi). This test shall be conducted at room temperature. An overpressure of 200 psi shall be applied to the transmitter for 10 minutes. Not less than 1 hour following this test, the transmitter shall be subjected to the scale error test on increasing pressure only. The scale error shall not exceed the tolerances specified in table I for after overpressure. Application of the 200-psi overpressure shall position the pointer of the indicating standard off scale approximately midway between the 100- and 0-psi graduations as required for the overpressure stop specified in 3.4.1.8.

4.6.21.2 Overpressure (500 psi). This test shall be conducted at room temperature with the case immersed in water and only one sample transmitter shall be used. The case shall be punctured or the seal shall be broken and a pressure of 500 psi shall be applied at the input pressure port. Leakage of the internal mechanism through the hole in the case shall be cause for rejection. The sample transmitter need not

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be checked for accuracy following this test. The transmitter shall then be disassembled and inspected for compliance with 3.4.1.7.1, 3.4.1.7.2, 3.4.1.9, and 3.4.1.9.1.

4.6.22 Mounting torque test. The mounting torque test shall be conducted at room temperature. One transmitter shall be mounted on suitable mounting pad. An unbalanced force of at least 40 pounds shall be applied to the 2.062 hex nut on the transmitter in such a manner as to generate a torque of 720 inch-pounds for a period of 2 minutes. The force and torque shall be applied in such a manner as to simulate installation or removal of the transmitter with a standard torque wrench. At the conclusion of this test, the transmitter shall be subjected to the scale error test. Distortion of the case or failure to meet the room temperature scale error requirements shall be cause for rejection.

4.6.23 Damping test. This test shall be conducted with MIL-L-7808 oil applied at the pressure port. A pressure of 100 psi shall be applied to the transmitter. A quick release valve shall be used to release the pressure. The time for the indication to change from 100 to 10 psi shall be 4.0±2 seconds at a temperature of -40±2°C. The test shall be repeated at a temperature of 177±2°C. The time for the indication to change from 100 to 10 psi shall be 0.1 to 1.0 seconds. A similar test shall be performed by raising the pressure from 0 to 50 psi within 2.0 seconds at a temperature of -40± 2°C. The transmitter indication shall rise to at least 40 psi within 22 seconds.

4.6.24 Helium test. When the transmitter is hermetically sealed, the case shall be punctured or the filling tube cut and the case tested by means of a mass-spectrometer-type helium leak detector. Failure to detect helium leakage shall be cause for rejection.

4.6.25 Reliability tests.

4.6.25.1 Test program. Prior to starting the reliability test, the contractor shall prepare a detailed reliability test program in accordance with MIL-STD-781 and submit it to the contracting activity for formal approval.

4.6.25.2 Reliability qualification test. The reliability test shall be conducted to demonstrate an MTBF of 2000 hours at a confidence level of 90% (decision risk = 10%; discrimination ratio = 1.5: θ_0 = 3000 hours; θ_1 = 2000 hours).

4.6.25.2.1 Test cycle. A minimum of ten TRU-80/A Oil Pressure Transmitters shall accumulate test time by being subjected to a series of environmental test cycles. Each environmental test cycle shall be in accordance with Figure 5 and the following:

Power: Off = 0 volts (15 minutes during -54°C and +177°C, 30 minutes during 25°C)

On = 26 ± 2 VAC @ 400 ± 5Hz

Oil Line Pressure: Off = 0 psi (15 minutes during -54°C and +177°C)
(Using a suitable liquid)

Pulse = Oscillate at 60 cycles per minute (cpm) from 35 to 50 psi ± 3 psi

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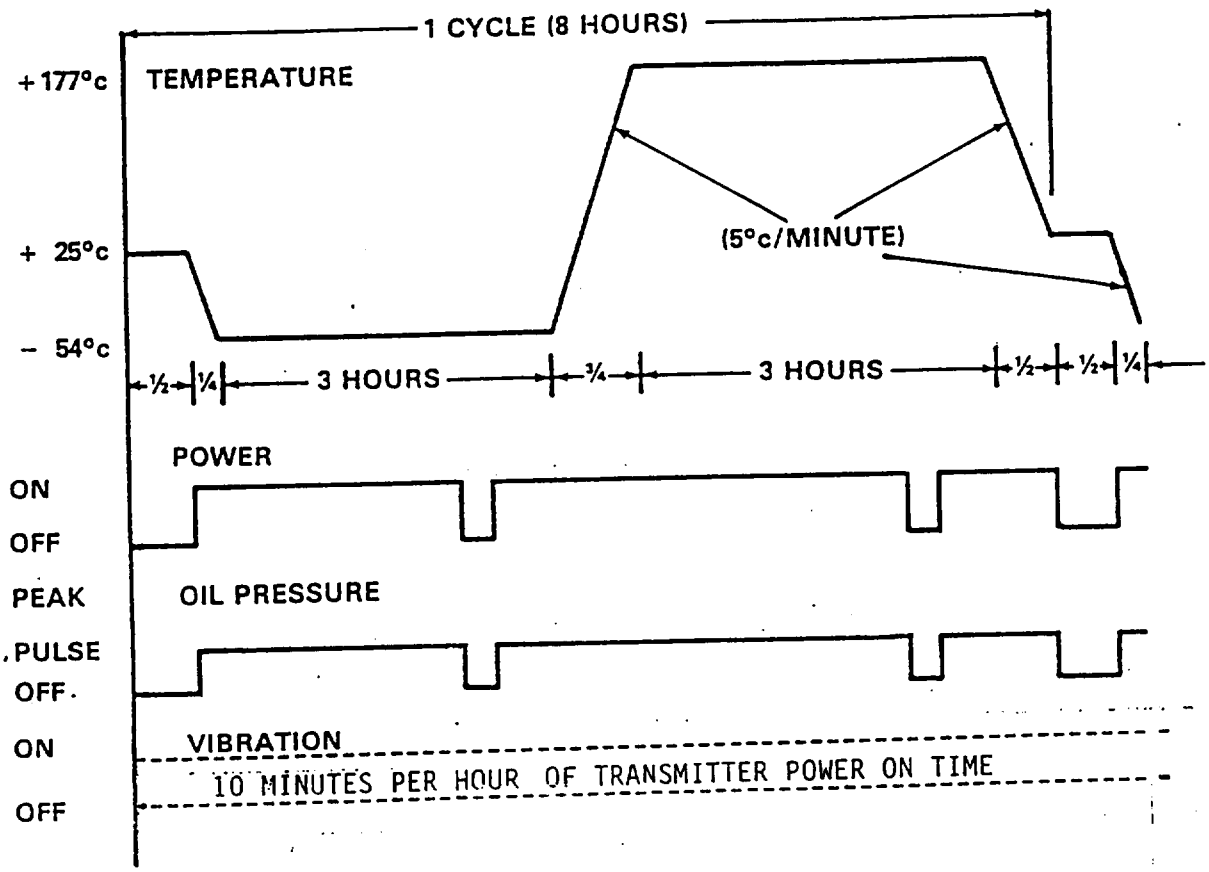


FIGURE 5. Environmental test cycle.

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Vibration: Off = none

On = Single axis random vibration per Figure 4 for 10 minutes per hour. Each 30 test cycles the transmitter shall be reoriented 90° so that another axis is tested.

One complete environmental test cycle (8 hours) shall equal seven hours reliability time.

4.6.25.2.2 Reliability test durations. Once every 30 test cycles each transmitter shall be subjected to and shall pass the scale error test per paragraph 4.6.2 and Table 1 for (AFTER) reliability. The accept/reject reliability test time of the transmitters and their total number of failures.

4.6.25.3 Production sampling (for Army or Navy contracts only). Sample transmitters shall be subjected to a production reliability test in accordance with MIL-STD-781, test level F, except the upper temperature level shall be 145°C and the vibration frequency cycle shall be from 5 to 2,000 Hz and return during each 10-minute vibration period in accordance with MIL-STD-810, curve T, figure 514.1-5. Accept-reject criteria shall in accordance with test plan II. This test shall be conducted on at least one sample from each lot. Transmitters produced during 1 month shall constitute a lot. The number of samples shall be as specified by the contracting activity. Testing shall be started with the first months production and the samples shall be tested until an accept or reject decision is reached. Unless otherwise specified in the contract or order, no transmitters shall be shipped until an accept decision is reached for each lot.

4.6.26 Sensors matching. The transmitter adjustment shall be adjusted, and locked, for operation with the test indicator and the adjustment shall not be further disturbed for the balance of the "Sensor Matching" test. The transmitter shall be tested at room temperature at the test points specified in table I for sensors matching with pressures applied per paragraphs 4.6.26.1 and 4.6.26.2. Both up-scale and down-scale readings shall be taken before and after tapping.

4.6.26.1 20 PSI vent pressure. A 20 PSI pressure shall be applied to the vent connection of the transmitter. Pressures shall be applied to the pressure connection which shall exceed that applied to the vent connection which shall exceed that applied to the vent connection by each of the test points listed in table I.

4.6.26.2 40 PSI vent pressure. A 40 PSI pressure shall be applied to the vent connection of the transmitter. Pressures shall be applied to the pressure connection which shall exceed that applied to the vent connection by each of the test points listed in table I.

4.7 Inspection and testing of preparation for delivery. Preparation for delivery shall be tested in accordance with MIL-P-116 to insure compliance with section 5 of this specification.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073/1A, as specified (see 6.2)

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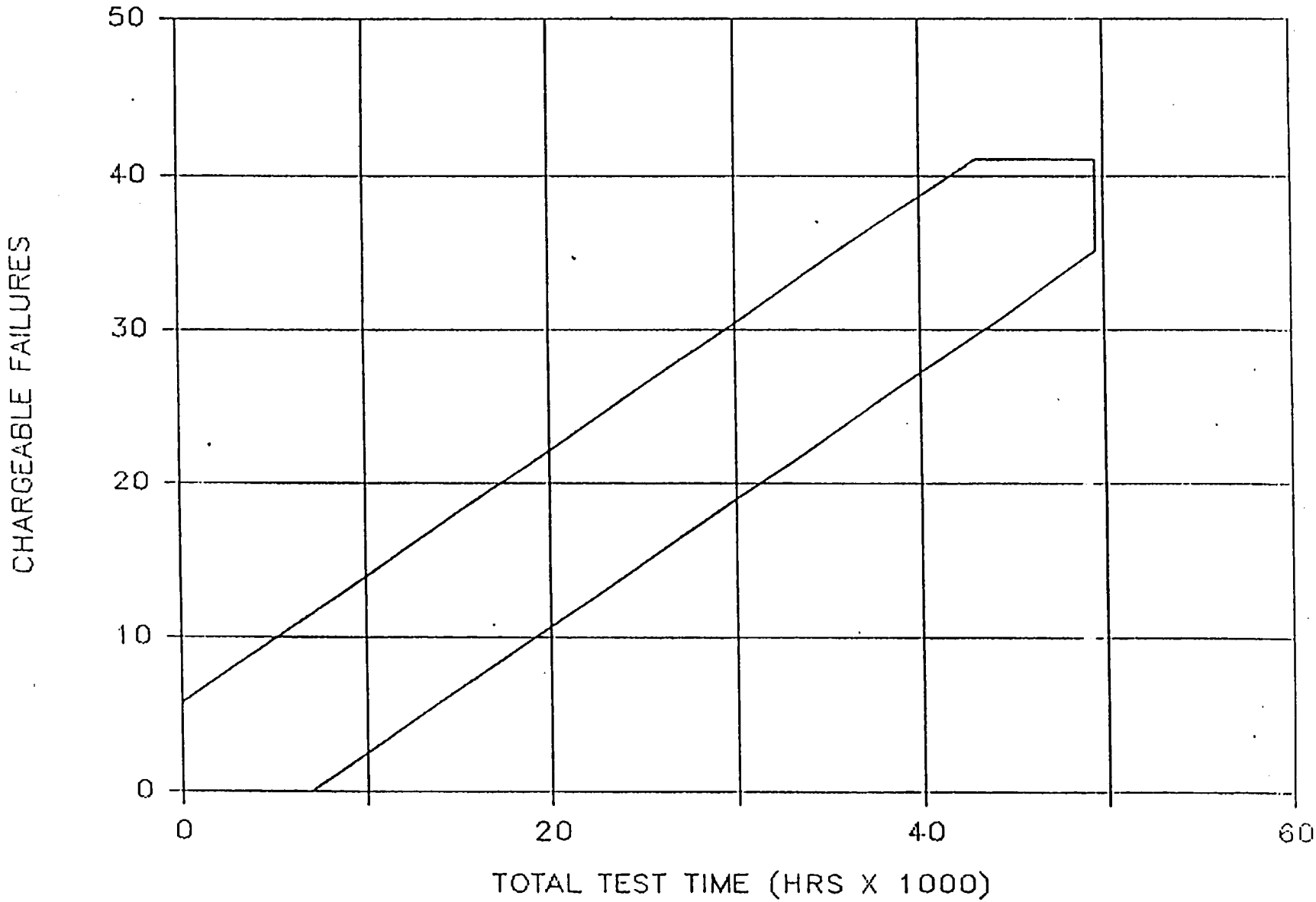


FIGURE 6. Reliability accept/reject curve.

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5.1.1 Level A.

5.1.1.1 Cleaning. Transmitter shall be cleaned in accordance with process C-I of MIL-P-116.

5.1.1.2 Drying. Transmitter shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation Application. Preservative shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each transmitter shall be packaged in quantity unit packs of one each in accordance with Method IC-I of MIL-P-116. Each transmitter shall be placed in a PPP-B-636 fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the Transmitter when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each transmitter shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The industrial preservation of the transmitter shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Transmitter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as minimum tare and cube consistent with the protection required.

5.2.2 Level B. Transmitter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use.

5.2.4 Industrial. The packaged transmitter shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The TRU-80/A transmitter is intended for use in transmitting, by electrical signal, the oil pressure in the lubricating system of aircraft engines to an indicator or indicators.

6.2 Acquisition requirements. Acquisition documents must specify the following:

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a. Title, number, and date of this specification.

b. Information, if any, to be submitted with the qualification samples.

c. Levels of packaging and packing required, special shipment marking, and reinspection date marking (see section 5).

d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Oklahoma City Air Logistics Center/MMEOR, Tinker AFB, OK 73145-5990.

6.4 Definitions. For the purpose of this specification, the following definitions will apply:

6.4.1 Good as new. Good as new means the transmitter has been operated less than 10 percent of the specified MTBF operation.

6.4.2 Refurbished. Refurbished means that the transmitter has been completely overhauled with all component parts meeting current part standards, and the transmitter has been subjected to and met all the requirements for a new transmitter.

6.5 Subject term (key word) listing.

Differential
Synchro Style
25 GRMS

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensivenss of the changes.

Custodians:
AIR FORCE -99
NAVY -AS

Preparing Activity:
AIR FORCE -71

Project number:
6620-0477

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
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4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
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PREPARING ACTIVITY	
a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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